

Structured Modal-Energy Dissipation for Inter-Area Oscillations in Reduced Low-Inertia Power Systems

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Abstract

The replacement of synchronous generation by inverter-interfaced distributed energy resources (DERs) lowers system inertia and leaves inter-area oscillations weakly damped, while aggregate frequency response is increasingly set by a deliberate allocation of virtual inertia and damping. Such allocation can be designed around the center-of-inertia (COI) response. Any supplementary wide-area damping action must not disturb it. This paper introduces a controller handling inter-area oscillations in reduced low-inertia systems. Through an internet communication channel the controller feeds back the selectively measured nodal frequencies and injects aggregate-neutral proportional damping. It adds and subtracts power distributed over the participating buses. To preserve the COI behaviour, the control action is chosen such that its net injected power sums to zero at every instant. By the initial- and final-value theorems, the initial COI rate of change of frequency (RoCoF) and the steady-state COI frequency endpoint are proven invariant. Numerically, the COI frequency nadir is left essentially unchanged. We test on an augmented twelve-bus Kundur benchmark, evaluated with both linear and nonlinear models. The controller tightens the inter-area frequency spread around the COI while preserving the designed COI frequency response. The same behaviour holds on the WECC, SC500 and Texas2000 systems for multiple amounts of participating damping nodes.

1 Introduction

The transition to renewable generation is replacing synchronous generators (SGs) with inverter-interfaced distributed energy resources (DERs) [1]. In Germany renewables already supply more than half of the electricity generated [2]. Germany's network regulator identifies the resulting loss of rotational inertia as a threat to secure operation [3, 4]. Falling inertia accelerates the post-disturbance frequency dynamics and erodes the rate-of-change-of-frequency (RoCoF) and nadir margins on which the grid depends [5, 6]. Novel grid codes require converters to provide grid-forming behaviour and instantaneous reserve [7]. At the same time, the reduced damping of converter-dominated networks leaves the inter-area electromechanical modes more lightly damped [8]. These modes are the slow oscillations of groups of units that swing coherently in frequency, against one another [9–11].

Damping these modes has long relied on wide-area control, which feeds back remote frequency or angle measurements to coordinate the response across areas [12, 13]. Phasor-measurement and converter-based realisations now damp inter-area modes with sparse communication and limited model information [14, 15].

In parallel, the aggregate response of low-inertia systems is increasingly set by the deliberate allocation of virtual inertia and damping to the DERs [8, 16, 17]. The hybrid allocation of [18] tunes these parameters so the system meets both small-signal and frequency-stability criteria, expressed on the center-of-inertia (COI) response. A supplementary damping

controller placed on top of such an allocation must not undo the COI behaviour it was designed for. This coupling between a wide-area damping action and the carefully shaped aggregate response has received little attention.

We propose structured modal-energy dissipation, realised as a zero-sum damping controller. On a Kron-reduced network model [19] the controller feeds back the measured frequencies. It adds and subtracts power among the participating DERs to dissipate the inter-area spread about the COI.

We call the dissipation structured as we limit it to the inter-area modal coordinates, the directions orthogonal to the aggregate. Further it only needs selective communication and partial measurement. We prove that the initial COI RoCoF and the steady-state endpoint are invariant under our approach. In contrast, the small-signal placement of the allocation [18] is not preserved and is re-checked directly from the closed-loop eigenvalues. The frequency nadir is not invariant. As no convex surrogate for it is presented, the nadir is verified numerically. We demonstrate the scheme on an augmented twelve-bus Kundur benchmark on both linear and nonlinear models. The behaviour then persists on the WECC, SC500 and Texas2000 systems under a participation sweep.

2 Small-Signal Model

We introduce the model used for the small-signal stability analysis. On the Kron-reduced transmission network it reads

$$\begin{aligned} \dot{\theta}(t) &= \omega(t), \\ M\dot{\omega}(t) + D\omega(t) + L\theta(t) &= P^{\text{dem}}(t) + S^g P^m(t) + P^K(t), \end{aligned} \quad (1)$$

where $\theta(t), \omega(t) \in \mathbb{R}^n$ are the vectors of bus voltage-angle and frequency deviations from a fixed operating point, $M = \text{diag}(m_i) \succ 0$ and $D = \text{diag}(d_i) \succeq 0$ are the diagonal inertia and damping allocations, and $L = L^\top \succeq 0$ is the reduced network matrix further explained in Section 2.1.

The right-hand side collects three active-power contributions. The vector $P^{\text{dem}}(t)$ gathers the demanded (exogenous) power at the nodes, including load disturbances. The vector $P^m(t)$ is the mechanical/governor power, placed at its buses by the selection matrix S^g of Section 3. It is treated as constant for the small-signal analysis and is later equipped with slower dynamics in Section 3 to capture the endpoint and nadir behaviour. The vector $P^K(t)$ denotes the power injected by the zero-sum damping controller designed in Section 4. It has either both positive and negative entries at once or is zero, such that the total power sums to zero at every instant.

The diagonal entries of M and D are obtained from system analysis. For SGs and grid-following (GFL) DERs they follow from linearisation, regression or other identification procedures. The precise choice is left to the user. For grid-forming (GFM) DERs they correspond to the allocated virtual inertia and damping. The diagonal submatrices M_{der} and D_{der} collect the entries of M and D at the allocatable DER nodes, excluding the SGs. Throughout we adopt the inertia-damping allocation of [18] as the baseline that supplies M_{der} and D_{der} . We make this separation due to different time scales of the SGs and DERs behaviour.

2.1 Reduced network matrix

Starting from the lossless active power-flow equation,

$$P_i = \sum_{j \in \mathcal{N}_i} V_i V_j B_{ij} \sin(\theta_i - \theta_j), \quad (2)$$

where P_i is the active power injected from bus i , $\mathcal{N}_i$ the set of buses adjacent to i , V_i, V_j the voltage magnitudes, B_{ij} the line susceptance and $\theta_i - \theta_j$ the angle difference across line (i, j) . Around a fixed operating point (V^*, θ^*) the voltage magnitudes are treated as constants (close to 1 per unit) and (2) is linearised with respect to the angle deviations. We then eliminate the load buses, as they carry no states and both constant-power and ohmic loads can be written algebraically. Analogous to [18] this leaves the reduced network matrix L as a Kron-reduced power-flow Jacobian [19]. Under the lossless-network, fixed-voltage and small-angle assumptions it has the structure of a weighted graph Laplacian: it captures how generator-bus power injections change with generator-bus angle deviations after the load-bus variables have been eliminated. In particular $L\mathbf{1} = 0$ and $\mathbf{1}^\top L = 0^\top$, where $\mathbf{1} = [1, 1, \dots, 1]^\top$. For brevity we refer to [18] for its derivation.

2.2 Center of inertia

Allocation of virtual inertia for DERs, as well as conventional planning, is concerned with the behaviour of the grid's center of inertia (COI) rather than with each individual node. Working with the single COI frequency, instead of the n nodal frequencies, reduces the number of imposed conditions. The hybrid

allocation of [18] likewise targets the aggregate COI response rather than nodal guarantees, forming an aggregate differential equation by summing the diagonal entries of the inertia and damping matrices.

We adopt the same aggregate viewpoint. As we will be introducing non-diagonal damping we define the total inertia and the COI frequency

$$m_\Sigma := \mathbf{1}^\top M \mathbf{1} = \sum_i m_i > 0, \quad \omega_{\text{COI}}(t) := \frac{\mathbf{1}^\top M \omega(t)}{m_\Sigma}, \quad (3)$$

together with the total damping $d_\Sigma := \mathbf{1}^\top D \mathbf{1}$. Every aggregate quantity is scalar e.g. the COI power terms $p_\Sigma^{\text{dem}}(t) = \mathbf{1}^\top P^{\text{dem}}(t)$, $p_\Sigma^m(t) = \mathbf{1}^\top S^g P^m(t)$ and $p_\Sigma^K(t) = \mathbf{1}^\top P^K(t)$. The COI dynamics read

$$m_\Sigma \dot{\omega}_{\text{COI}}(t) + d_\Sigma \omega_{\text{COI}}(t) = p_\Sigma^{\text{dem}}(t) + p_\Sigma^m(t) + p_\Sigma^K(t). \quad (4)$$

3 Governor Model

The small-signal model of Section 2 assumes constant mechanical power $P^m(t)$ to capture the oscillatory small-signal behaviour. It cannot reproduce the frequency nadir or the steady-state endpoint, both of which are set by the slower primary droop response whose mechanical power ramps with a finite lag. In practice $P^m(t)$ follows the slower dynamics of a governor at the buses of the SGs, whose mechanical power does not respond instantaneously.

3.1 Nodal governor: The governed buses, indexed by the set $\mathcal{G}$, carry the mechanical powers stacked in $P^m(t) \in \mathbb{R}^{|\mathcal{G}|}$. Each entry follows a first-order turbine lag driven by its local frequency,

$$T^g \dot{P}^m(t) = -P^m(t) - R^{-1}(S^g)^\top \omega(t), \quad (5)$$

with turbine time constants $T^g = \text{diag}(\tau_i) \succ 0$, droop gains $R^{-1} = \text{diag}(r_i^{-1}) \succeq 0$ and a selection matrix $S^g \in \mathbb{R}^{n \times |\mathcal{G}|}$ that places each governor at its respective bus. The contribution entering the swing dynamics (1) is accordingly $S^g P^m(t)$, which is zero at every ungoverned bus.

3.2 Reduced COI governor: The COI analysis works with the single aggregate frequency $\omega_{\text{COI}}(t)$ rather than the n nodal frequencies, so the nodal governor must be reduced to a matching scalar law. Aggregating (5) over the governed buses replaces the constant aggregate mechanical power p_Σ^m of (4) by an equivalent lumped governor state,

$$\dot{p}_\Sigma^m(t) = \frac{1}{\tau} (-p_\Sigma^m(t) - r^{-1} \omega_{\text{COI}}(t)), \quad (6)$$

with equivalent turbine time constant τ and equivalent droop coefficient r . The pair (4) and (6) is the COI counterpart of the turbine-augmented swing dynamics used in the hybrid allocation [18].

4 Zero-Sum Damping P^K

In modern power systems we assume the availability of communication between distinct energy sources. Many residential solar systems, for instance, already host a telecommunications

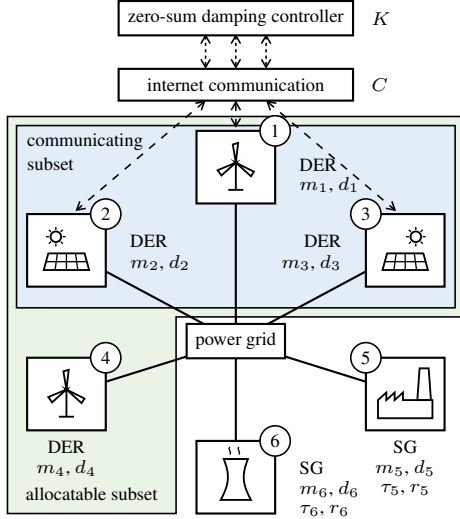


Fig. 1 Setting of the paper on a reduced network of inverter-interfaced DERs and SGs.

interface next to their grid connection. We leverage this availability without imposing the unrealistic condition that every unit can communicate with every other. Instead we assume *selective* communication availability for a subset of the DERs: solar farms that are already connected for telemetry and want to contribute to grid stability, or local home systems that improve stability by exchanging information with units in their geographic neighbourhood. We assume that only a subset of the entries of the frequency vector $\omega(t)$ can be measured and communicated.

Figure 1 sketches the resulting setting. One subset of the units carries the inertia and damping allocation. Inside it a smaller subset also reports its measured frequency and receives a damping injection signal in return. The remaining units, here the SGs, are left to their governors.

Throughout, the communication is taken to be ideal, that is instantaneous, lossless and continuously sampled. This is a strong assumption, so the present results are an idealised upper bound, with sampling, latency and packet loss left to future work.

4.1 Inter-area coordinates

A single channel matrix $C \in \mathbb{R}^{n \times r}$ encodes both the measurement restriction and the inter-area structure. It is required to be aggregate-neutral,

$$\mathbf{1}^\top C = 0, \quad (7)$$

and to have zero rows for every node whose frequency cannot be measured, so that $C^\top \omega(t)$ depends only on communicated quantities.

Consider three DERs sharing one line as sketched in Fig. 2. Their frequency is measured and communicated. We pick the aggregate-neutral projector used throughout,

$$C = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (8)$$

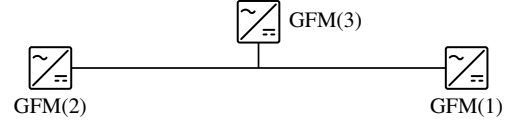


Fig. 2 Mini example: three units sharing one line. Every unit communicates with every other.

Each column sums to zero, so $\mathbf{1}^\top C = 0$ holds columnwise. The communicated coordinates are the deviations of the nodal frequencies from their mean,

$$C^\top \omega(t) = \frac{1}{3} \begin{bmatrix} 2\omega_1(t) - \omega_2(t) - \omega_3(t) \\ -\omega_1(t) + 2\omega_2(t) - \omega_3(t) \\ -\omega_1(t) - \omega_2(t) + 2\omega_3(t) \end{bmatrix}. \quad (9)$$

The first entry is the first DER swinging against the combined second and third, the second direction is the second DER against the combined first and third etc. Of the $n = 3$ nodal degrees of freedom one is the COI mode, which leaves $n - 1 = 2$ genuine relative directions.

4.2 Zero-sum control law

For damping the inter-area oscillations we use a proportional control law with gain $K \succeq 0$ and channel matrix C , giving

$$P^K(t) = -CKC^\top \omega(t) = -D^K \omega(t), \quad (10)$$

A superscript K names a quantity built from the gain, as D^K is here. A superscript 0 marks the same quantity without it, which we need only where a proof puts the two side by side. By (7) the aggregate of the injection $P^K(t)$ vanishes,

$$p_\Sigma^K(t) = -(\mathbf{1}^\top C) K C^\top \omega(t) = 0, \quad (11)$$

so that the zero-sum damping controller redistributes active power among the participating units without creating a net injection. Consequently the COI dynamics (4) reduce to

$$m_\Sigma \dot{\omega}_{\text{COI}}(t) + d_\Sigma \omega_{\text{COI}}(t) = p_\Sigma^{\text{dem}}(t) + p_\Sigma^m(t) + 0, \quad (12)$$

i.e. the zero-sum injection does not enter the *instantaneous* COI balance. It does, however, influence the COI state *trajectory*.

4.3 Closed-loop model

To assess the impact of the zero-sum damping on the overall network dynamics we formulate the closed-loop system, both in state space and as a transfer function. Throughout we consider load-step scenarios exclusively. Parametric-uncertainty scenarios and line trips fit the same framework but burden the notation.

The governor-augmented reduced model of Section 3, in its full nodal form and driven by a load step, is

$$M \dot{\omega}(t) = -D_{\text{res}}^K \omega(t) - L\theta(t) + B^p \bar{p} \sigma(t) + S^g P^m(t), \quad (13)$$

$$T^g \dot{P}^m(t) = -P^m(t) - R^{-1}(S^g)^\top \omega(t), \quad (14)$$

$$\dot{\theta}(t) = \omega(t), \quad (15)$$

The zero-sum injection $P^K(t) = -D^K \omega(t)$ of (10) merges with the native damping into the *effective damping*

$$D_{\text{res}}^K := D + D^K. \quad (16)$$

The load-step disturbance is $P^{\text{dem}}(t) = B^p \bar{p} \sigma(t)$ (σ the unit step, $\bar{p}$ the scalar step magnitude and $B^p \in \mathbb{R}^n$ its nodal distribution).

The deviation coordinates are stacked in the state $x(t) = [\theta(t)^\top \omega(t)^\top P^m(t)^\top]^\top$, which starts at the origin, $x(0) = 0$. With the COI frequency as output, (13)-(15) can be written as the state-space model

$$\dot{x}(t) = Ax(t) + Bu(t), \quad \omega_{\text{COI}}(t) = C_{\text{COI}}x(t), \quad (17)$$

with scalar input $u(t) = \bar{p}\sigma(t)$ and full-order matrices

$$A = \begin{bmatrix} 0 & I & 0 \\ -M^{-1}L & -M^{-1}D_{\text{res}}^K & M^{-1}S^g \\ 0 & -(T^g)^{-1}R^{-1}(S^g)^\top & -(T^g)^{-1} \end{bmatrix}, \quad (18)$$

$$B = \begin{bmatrix} 0 \\ M^{-1}B^p \\ 0 \end{bmatrix}, \quad C_{\text{COI}} = \begin{bmatrix} 0 & \frac{\mathbf{1}^\top M}{m_\Sigma} & 0 \end{bmatrix}. \quad (19)$$

The disturbance-to-COI transfer function is then

$$G_{\text{COI}}^K(s) = C_{\text{COI}}(sI - A)^{-1}B. \quad (20)$$

5 Impact of the Zero-Sum Damping on the COI Behaviour

In mixed-generation networks the COI behaviour is deliberately shaped through the allocation of virtual inertia and damping. When an overlaying scheme such as the zero-sum damping controller of Section 4 is added, this designed behaviour must be conserved. We therefore take the explicit allocation, that of [18], which selects the diagonals M and D with respect to small-signal stability and frequency stability and analyse how the zero-sum damping D^K affects each design criterion and whether additional measures are required.

5.1 Oscillation-damping (small-signal) constraints

For small-signal stability the allocation of [18] places the swing modes into a desired conic region by acting on the eigenvalues (EVs) of the ungoverned system matrix [18, Eq. (10)]. This placement concerns the DER electromechanical modes, so it is built on the DER inertia M_{der} and damping D_{der} , the diagonal matrices $\text{diag}(m_{i,\text{der}})$ and $\text{diag}(d_{i,\text{der}})$ collecting only the nodes of the allocatable subset of Fig. 1. With the zero-sum damping controller in place the corresponding partial state matrix reads

$$A_{\text{der}}^K = \begin{bmatrix} 0 & I \\ -M_{\text{der}}^{-1}L_{\text{der}} & -M_{\text{der}}^{-1}D_{\text{res,der}}^K \end{bmatrix}, \quad D_{\text{res,der}}^K = D_{\text{der}} + D_{\text{der}}^K. \quad (21)$$

It shares the swing structure of the state matrix A in (18), with the governor channel removed and every full-system operator replaced by its DER-node restriction. For the baseline case we take matrix A_{der}^0 , without the zero-sum damping controller. It is formed by the same expression as in (21) with index 0, that is with D_{der} in place of $D_{\text{res,der}}^K$. The placement conditions of [18, Thm. 1] are matrix inequalities in the inertia and the damping and use neither their diagonality nor their per-node structure. We hold the allocated M_{der} and D_{der} fixed, so the conditions stay linear matrix inequalities in K . The EVs of (21) can therefore again be placed in the desired conic region by the same convex program, now with $K \geq 0$ among the decision variables.

5.2 Frequency-stability constraints

For frequency stability the allocation [18] examines the system response over a longer horizon and therefore retains the turbine and governor dynamics. We accordingly work with the governor-augmented closed-loop model (13)-(15), with matrices (18)-(19) and COI transfer function (20), which can be rewritten as

$$G_{\text{COI}}^K(s) = \frac{\mathbf{1}^\top M}{m_\Sigma} Z^K(s)^{-1} B^p, \quad (22)$$

with the frequency-dependent matrix

$$Z^K(s) = sM + D_{\text{res}}^K + \frac{1}{s}L + G^{\text{gov}}(s), \quad (23)$$

$$G^{\text{gov}}(s) = S^g \text{diag}\left(\frac{r_i^{-1}}{\tau_i s + 1}\right) (S^g)^\top.$$

Let $G_{\text{COI}}^0(s)$ denote the transfer function with the baseline allocation without the zero-sum damping controller. The allocation checks three quantities on the COI response to a load step: the rate of change of frequency (RoCoF), the steady-state endpoint and the frequency nadir. As the inertias are allocated for the system poles to be in the left open half-plane we apply the following standing assumption.

Assumption 1. *All modes of the closed-loop matrix A of (18) except the rigid-rotation mode with eigenvector $r = [\mathbf{1}^\top \ 0^\top \ 0^\top]^\top$; ($Ar = 0$ since $L\mathbf{1} = 0$) are asymptotically stable. Under a step the frequencies then synchronise, $\omega(t) \rightarrow \omega_\infty \mathbf{1}$ and $P^m(t) \rightarrow P_\infty^m$ as $t \rightarrow \infty$.*

5.2.1 COI RoCoF (initial-value theorem, $s \rightarrow \infty$): The initial RoCoF is $\dot{\omega}_{\text{COI}}(0^+)$. With zero initial conditions $\mathcal{L}\{\dot{\omega}_{\text{COI}}\} = s\Omega_{\text{COI}}(s)$, and $\Omega_{\text{COI}}(s) = G_{\text{COI}}^K(s)\bar{p}/s$ by (20), the initial-value theorem gives

$$\dot{\omega}_{\text{COI}}(0^+) = \lim_{s \rightarrow \infty} s(s\Omega_{\text{COI}}(s)) = \lim_{s \rightarrow \infty} sG_{\text{COI}}^K(s)\bar{p}. \quad (24)$$

Theorem 1. *Under zero initial conditions,*

$$\dot{\omega}_{\text{COI}}(0^+) = C_{\text{COI}}B\bar{p} = \frac{\mathbf{1}^\top B^p}{m_\Sigma} \bar{p}, \quad (25)$$

independently of D_{res}^K . Thus the zero-sum damping controller leaves the initial RoCoF unchanged.

Proof: Expanding $s(sI - A)^{-1} = I + \frac{1}{s}A + \frac{1}{s^2}A^2 + \dots$ as a Neumann series yields the Markov expansion

$$sG_{\text{COI}}^K(s) = C_{\text{COI}}B + \frac{1}{s}C_{\text{COI}}AB + \frac{1}{s^2}C_{\text{COI}}A^2B + \dots \quad (26)$$

Every term beyond the first carries a factor $1/s$ and vanishes as $s \rightarrow \infty$. These are precisely the terms containing the state matrix A and hence all of L , D_{res}^K , S^g , T^g , R^{-1} . Only the direct feedthrough $C_{\text{COI}}B$ remains. The zero blocks of C_{COI} and B in (18)-(19) leave

$$C_{\text{COI}}B = \frac{\mathbf{1}^\top M}{m_\Sigma} M^{-1}B^p = \frac{\mathbf{1}^\top B^p}{m_\Sigma}, \quad (27)$$

Since the limit discards A entirely, the baseline G_{COI}^0 gives the same value as G_{COI}^K . $\square$

5.2.2 COI endpoint (final-value theorem, $s \rightarrow 0$): The endpoint is the steady COI frequency after the step. Because the angle coordinate is free to rotate, the COI output must be read in a form that is blind to the rigid mode. This is exactly

the projection in which the zero-sum damping controller also disappears.

Theorem 2. *Under Assumption 1,*

$$\lim_{t \rightarrow \infty} \omega_{\text{COI}}(t) = \frac{\mathbf{1}^\top B^p \bar{p}}{\mathbf{1}^\top D\mathbf{1} + \mathbf{1}^\top S^g R^{-1} (S^g)^\top \mathbf{1}} = \frac{\mathbf{1}^\top B^p \bar{p}}{\mathbf{1}^\top D\mathbf{1} + \sum_{i \in \mathcal{G}} r_i^{-1}}, \quad (28)$$

independently of L , of the turbine time constants T^g and of the zero-sum damping D^K . Consequently $G_{\text{COI}}^K(0) = G_{\text{COI}}^0(0)$.

Proof: By the final-value theorem,

$$\lim_{t \rightarrow \infty} \omega_{\text{COI}}(t) = \lim_{s \rightarrow 0} s \Omega_{\text{COI}}(s) = G_{\text{COI}}^K(0) \bar{p},$$

so it remains to evaluate the DC gain on $Y(s) := Z^K(s)^{-1} B^p$, the solution of $Z^K(s) Y(s) = B^p$ with $Z^K(s)$ in (23). By Assumption 1 every mode of A other than the rigid rotation r is asymptotically stable, and r has zero frequency component, so it is unobservable from ω . Hence Y has no pole at the origin and is analytic at $s = 0$ with $Y(0)$ finite. The limits in the two steps below may therefore be taken termwise, and the limit of the product $\mathbf{1}^\top Z^K(s) Y(s)$ splits into the product of the limits.

Direction. Multiplying $Z^K(s) Y(s) = B^p$ by s gives

$[s^2 M + s D_{\text{res}}^K + L + s G^{\text{gov}}(s)] Y(s) = s B^p$. Letting $s \rightarrow 0$ leaves $L Y(0) = 0$, so $Y(0) \in \ker L = \text{span}\{\mathbf{1}\}$ and $Y(0) = \omega_\infty \mathbf{1}$: the network forces a common steady frequency.

Value. Left-multiplying $Z^K(s) Y(s) = B^p$ by $\mathbf{1}^\top$ and expanding,

$$\mathbf{1}^\top Z^K(s) = \underbrace{s \mathbf{1}^\top M}_{\rightarrow 0^\top} + \underbrace{\mathbf{1}^\top D_{\text{res}}^K}_{= \mathbf{1}^\top D} + \underbrace{\frac{1}{s} \mathbf{1}^\top L}_{= 0^\top} + \mathbf{1}^\top G^{\text{gov}}(s). \quad (29)$$

We do two structural simplifications. First, the divergent term: since $\frac{1}{s} \mathbf{1}^\top L = 0^\top$ by $\mathbf{1}^\top L = 0^\top$, the $1/s$ singularity is annihilated. Second, the zero-sum damping D^K drops out of the effective damping, $\mathbf{1}^\top D_{\text{res}}^K = \mathbf{1}^\top D$, since $\mathbf{1}^\top D^K = 0^\top$ by the zero-sum property (11). As $s \rightarrow 0$ the inertia term vanishes and $G^{\text{gov}}(0) = S^g R^{-1} (S^g)^\top$. With $Y(0) = \omega_\infty \mathbf{1}$,

$$(\mathbf{1}^\top D\mathbf{1} + \mathbf{1}^\top S^g R^{-1} (S^g)^\top \mathbf{1}) \omega_\infty = \mathbf{1}^\top B^p. \quad (30)$$

Applying C_{COI} and using $\mathbf{1}^\top M\mathbf{1} = m_\Sigma$ gives

$$G_{\text{COI}}^K(0) = \frac{\mathbf{1}^\top M}{m_\Sigma} \omega_\infty \mathbf{1} = \omega_\infty, \quad (31)$$

hence (28). Since $(S^g)^\top \mathbf{1}$ is the all-ones vector over the generator set $\mathcal{G}$, $\mathbf{1}^\top S^g R^{-1} (S^g)^\top \mathbf{1} = \sum_{i \in \mathcal{G}} r_i^{-1}$. The denominator contains neither D^K nor T^g , so $G_{\text{COI}}^K(0) = G_{\text{COI}}^0(0)$. $\square$

5.2.3 Frequency nadir: The COI output does not see the injected damping directly: D^K is zero-sum ($\mathbf{1}^\top P^K(t) = 0$) and drops out of the aggregate balance (12). Recall the Markov expansion (26) used to prove Theorem 1. At $s \rightarrow \infty$ only the leading term $C_{\text{COI}} B$ stays relevant, as it is independent of A and hence of D^K . For finite s the higher-order terms enter. These involve the closed-loop matrix A of (18), which depends on D^K through D_{res}^K . The full COI transfer function (22) and the resulting state trajectory are therefore modified by the zero-sum damping controller.

Since the inertia and damping diagonals are designed before the zero-sum damping controller is added, we check the grid-code nadir condition on the resulting closed-loop response as an a posteriori certificate on a proposed gain K . This check can be done analytical but not naturally convex in K . Therefore it

does not fit naturally into the convex optimisation framework of [18].

6 Case study: linear and nonlinear models

We test the proposed approach on the same examples considered in [18], for which we adjusted the reference code¹ to expose the zero-sum damping controller. This section revisits the augmented Kundur benchmark of [18] on both the linear aggregate model and the full nonlinear simulation. Section 7 then turns to a scalability study on the larger WECC, SC500 and Texas2000 networks.

Throughout the case studies we report frequencies in Hertz rather than the angular frequency deviation ω of the model, as is conventional in power-system analysis. The frequencies follow from the model state through $f_i(t) = f_0 + \omega_i(t)/(2\pi)$ with nominal frequency $f_0 = 60$ Hz.

6.1 Test-case setup

The augmented Kundur benchmark of [18] extends the classical Kundur two-area system into a three-area, twelve-bus network with two SGs, six grid-forming and three grid-following converters. After Kron reduction of the load buses the model retains nine inverter buses, each carrying one nodal frequency $f_i(t)$. The disturbance is a step load increase and the network topology is reproduced in Fig. 3. The allocated inertias and damping diagonals are taken as given from the reference allocation of [18], while the graph structure (the Laplacian after Kron reduction) is extracted from the Simulink network model. All remaining numerical parameters are listed in the adapted reference code. The conic design region is that of [18, Thm. 1], a decay margin β with $\text{Re}(\lambda) \leq -\beta$ and a cone half-angle ζ with $\sin \zeta \text{Re}(\lambda) + \cos \zeta |\text{Im}(\lambda)| \leq 0$, so the damping ratio is at least $\cos \zeta$. We keep $\beta = 3$ rad/s and $\cos \zeta = 0.1$ throughout.

The linear model has known weaknesses. The extremum of the COI RoCoF trajectory, for one, has opposite sign in the two models. The nonlinear response nevertheless stays within the analytic nadir and RoCoF bounds throughout, so we treat the aggregate model as a conservative worst case, verified on the nonlinear system later in this section, where the differences become apparent in the explicit COI trajectories.

6.2 Controller selection

Before applying the controller we check the stability and performance constraints of Section 5. We pick the full-communication, aggregate-neutral channel matrix

$$C = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top, \quad (32)$$

which is the symmetric idempotent projector onto the zero-mean subspace, with gain $K = 20 I$. Both are picked arbitrarily, K large enough to make the zero-sum damping controller clearly visible. Optimal design of C and K is left for future work.

¹<https://github.com/stkotum/Hybrid-Oscillation-Damping-and-Inertia-Management-for-DERS>

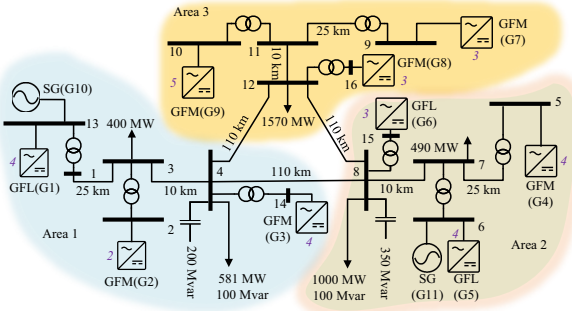
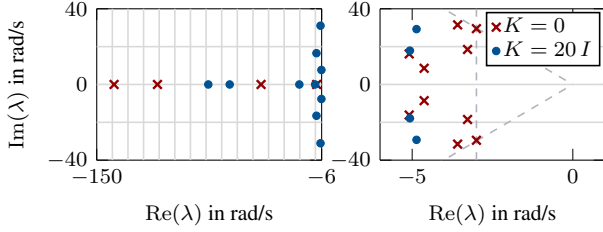


Fig. 3. Augmented Kundur test case, from [18]. © 2025 IEEE.


 Fig. 4. Twelve-bus example small-signal spectrum including the conic design region in dashed lines. Baseline A_{der}^0 ($K = 0$) compared to A_{der}^{20} ($K = 20I$). The right plot magnifies the region of the left one near the imaginary axis.

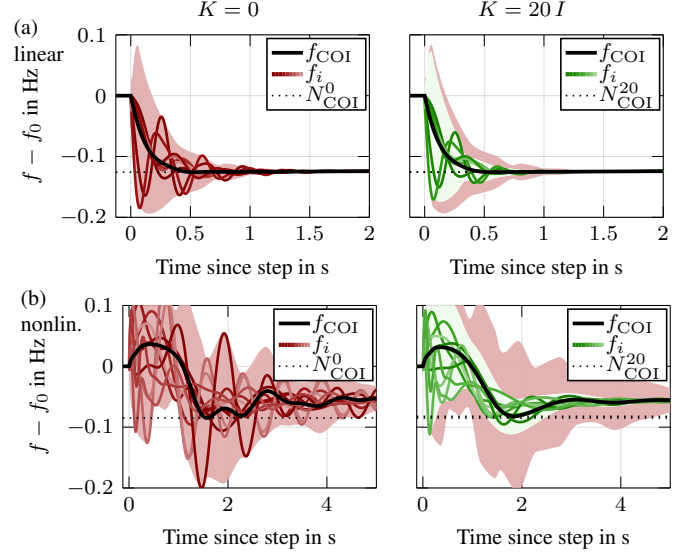
6.3 Oscillation-damping (small-signal) constraints

Figure 4 shows the inverter-bus EVs of the small-signal matrix (21) for the baseline allocation A_{der}^0 ($K = 0$) and with the zero-sum damping controller in place, A_{der}^{20} ($K = 20I$). The rigid mode at the origin and the synchronous-generator EVs are not displayed, as they influence neither the choice of virtual inertias nor the controller. With the zero-sum damping applied, the remaining EVs are pulled closer together. The EVs close to the origin move further into the left half-plane, while the ones with the smallest real part move towards the imaginary axis. This phenomenon is example-specific and does not hold in general.

6.4 Frequency-stability constraints

Theorems 1 and 2 of Section 5 establish that the initial COI RoCoF and the COI endpoint are invariant under the zero-sum damping controller, so neither needs to be re-checked here. The nadir, by contrast, depends on the finite-time closed-loop response and is evaluated numerically. For the twelve-bus example the baseline and controlled COI nadirs are $N_{\text{COI}}^0 \approx -0.12583$ Hz at $t \approx 0.528$ s and $N_{\text{COI}}^{20} \approx -0.12575$ Hz at $t \approx 0.560$ s. The zero-sum damping controller makes the dip marginally shallower (by about 0.08 mHz, i.e. 0.07%) and delays it by roughly 30 ms, leaving the nadir essentially unchanged.

Figure 5(a) reports the per-node frequency $f_i(t)$ relative to the nominal $f_0 = 60$ Hz in the linearised system, with the zero-sum damping controller off ($K = 0$) and on ($K = 20I$). The envelope over all $f_i(t) - f_0$ tightens the individual bus frequencies about the COI. The inter-area component of $f_i(t) - f_{\text{COI}}(t)$


 Fig. 5. Twelve-bus per-node frequency deviation, comparing controller off ($K = 0$, left) versus on ($K = 20I$, right) as well as (a) linearised model and (b) nonlinear simulation.

is markedly reduced once the controller is engaged, whereas $f_{\text{COI}}(t)$ stays nearly unchanged.

Figure 5(b) repeats the comparison on the nonlinear simulation. As anticipated in Section 6.1, the nonlinear response differs from the linear one: the uncontrolled case shows considerably more oscillation and a different COI trajectory. This is also why we make no claim of invariance for the full COI trajectory, the linear model already being an imprecise representation of the real system. The decrease in the inter-area modes under the zero-sum damping controller is nevertheless even larger than in the linear case. It has the same effect: it suppresses the inter-area swing of the nodal frequencies $f_i(t)$ while leaving $f_{\text{COI}}(t)$ essentially unchanged.²

7 Case study: scalability

The twelve-bus example of Section 6 established the concept but cannot show whether the behaviour is a property of the zero-sum damping controller or an artefact of a small benchmark. We therefore test scalability on WECC, the South Carolina 500-bus system (SC500) and the Texas 2000-bus system (Texas2000), each at full inverter penetration: every SG becomes an inverter-based DER, so the system carries no synchronous inertia. Each network is reduced to its inverter buses through the Kron-reduced power-flow Jacobian L of Section 2.1, so the small-signal matrix (21) and the reduced COI transfer function (22) keep the structure used throughout. We sweep the participation share ρ of DERs in the zero-sum damping from 0% to 100%, since full participation is unrealistic with a heterogeneous generation mix and only partially deployed wide-area communication.

²The nonlinear COI uses the same inertia-weighted definition $\mathbf{1}^\top M / m_\Sigma$ as the linear model.

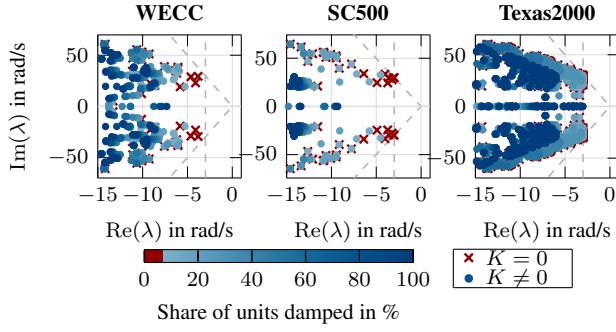


Fig. 6 Small-signal EV cluster across the activation sweep on WECC, SC500 and Texas2000.

7.1 Test-case setup

The cases follow [18]. The disturbance is a load step with nodal distribution B^p proportional to the initial per-bus demand, detailed in the adapted reference code linked in Section 6. The explicit values for M and D are published in our code, following the idea of allocation from [18]. This allocation places the baseline poles of A_{der}^0 , the matrix (21) at $K = 0$, on or inside the conic region and meets the frequency-stability constraints of Section 5.2, so the base case is conic-feasible by construction. The large cases carry no turbine-governor stage, so the droop total $\sum_{i \in G} r_i^{-1}$ in (28) is zero and the steady-state offset is set by the native damping total $1^T D 1$ alone.

The communicating subset is floored at two units whenever $\rho > 0$, so the zero-sum damping controller always has enough participants to act on. It carries the aggregate-neutral channel of (32) at every share.

7.2 Controller selection

As in Section 6 the controller uses a scalar gain $K = \kappa I$, one κ per case, $\kappa_{\text{WECC}} = 15$, $\kappa_{\text{SC500}} = 9$ and $\kappa_{\text{Texas2000}} = 4$, each tuned to harden the closed-loop frequency envelope while keeping every non-rigid EV of (21) inside the sector. For WECC and SC500 the spectrum sits deep in the interior, so κ can be large, whereas Texas2000 is sector-tight and κ is kept small. A non-diagonal K matched to the modal structure could improve the result, but a scalar gain isolates the effect of participation from controller shaping, so we leave a structured K to future work.

7.3 Oscillation-damping (small-signal) constraints

We treat the oscillation-damping behaviour first, through the spectrum of the closed-loop small-signal matrix A_{der}^K in (21) over the sweep. Figure 6 shows the EVs for the three cases, with the baseline EVs at $K = 0$ in red and the EVs at the activated shares shaded from light ($> 0\%$) to dark (100%) blue, against the sector boundary and damping cone.

The trend reproduces the twelve-bus observation at scale. As the share grows the swing EVs migrate left into the sector with little change in their imaginary parts. The WECC and SC500 clusters move well clear of the boundary, with rightmost non-rigid EV at $\text{Re}(\lambda) = -7.31$ rad/s and $\text{Re}(\lambda) = -9.30$ rad/s respectively. For Texas2000 the migration is more modest, with

the oscillatory modes pulled inwards while the binding real mode, pinned at $\text{Re}(\lambda) = -\beta$ by the baseline allocation, stays just inside at $\text{Re}(\lambda) = -3.006$ rad/s. In every case the closed-loop spectrum stays by design in the conic region across the full sweep, certified by the a posteriori sector check of Section 5.1 at every participation level.

7.4 Frequency-stability constraints

The initial RoCoF and the steady-state endpoint are invariant under the zero-sum damping controller by Theorems 1 and 2 so neither is re-derived here. The nadir is the only remaining quantity, a finite-frequency effect of the full networked transfer function (22) (Section 5.2.3) that we check numerically.

Figure 7 shows the result for the load step, shaded by share. The top row confirms the invariants visually, the 0% and participating $\Delta f_{\text{COI}}(t)$ traces sharing one initial slope and one endpoint. Their transient shape differs by case, with SC500 showing a genuine transient nadir near $t = 0.12$ s while the damping-dominated WECC and Texas2000 are essentially monotone, so their minimum coincides with the endpoint. The zero-sum damping controller again tightens the inter-area envelope while leaving the proven COI invariants fixed, so the effect is not an artefact of the small benchmark. The top row gives the COI dip, -0.044 , -0.020 and -0.025 Hz on WECC, SC500 and Texas2000, moving by less than 0.8 mHz across the sweep. The middle row gives the largest nodal deviation from the COI, $\max_i |f_i(t) - f_{\text{COI}}(t)|$, since every nodal frequency at every share would crowd the panel. It peaks at 0.20, 0.057 and 0.17 Hz, several times that dip. Those curves overlap within a line width, so the bottom row subtracts the 0% baseline. The reduction there reaches 0.034, 0.027 and 0.033 Hz, the same order as the dip itself. The change is not gradual. Adding particular channels moves the envelope in steps, so which DERs participate matters more than how many.

8 Conclusion and outlook

This paper introduced a zero-sum damping controller built on the aggregate-neutral channel matrix C . By Theorems 1 and 2 this controller injects inter-area damping while leaving the initial COI RoCoF and steady-state endpoint invariant and the nadir essentially unchanged. It was validated on a twelve-bus benchmark in linear and nonlinear scenarios as well as on the WECC, SC500 and Texas2000 systems under a partial participation sweep. The results are promising throughout and most pronounced in the nonlinear simulation, where the inter-area spread narrows even further than the linear model predicts.

Three directions remain. First, the hand-tuned scalar gain K should instead be posed as an optimisation comparing diagonal against non-diagonal K , ideally under a convex, or near-convex, nadir criterion so it fits the convex allocation framework. Second, internet communication effects must be studied, including signal sampling, the transmitted quantities and latency, for the scheme to be practical. Third, handling actuator limits and the availability of power must be addressed.

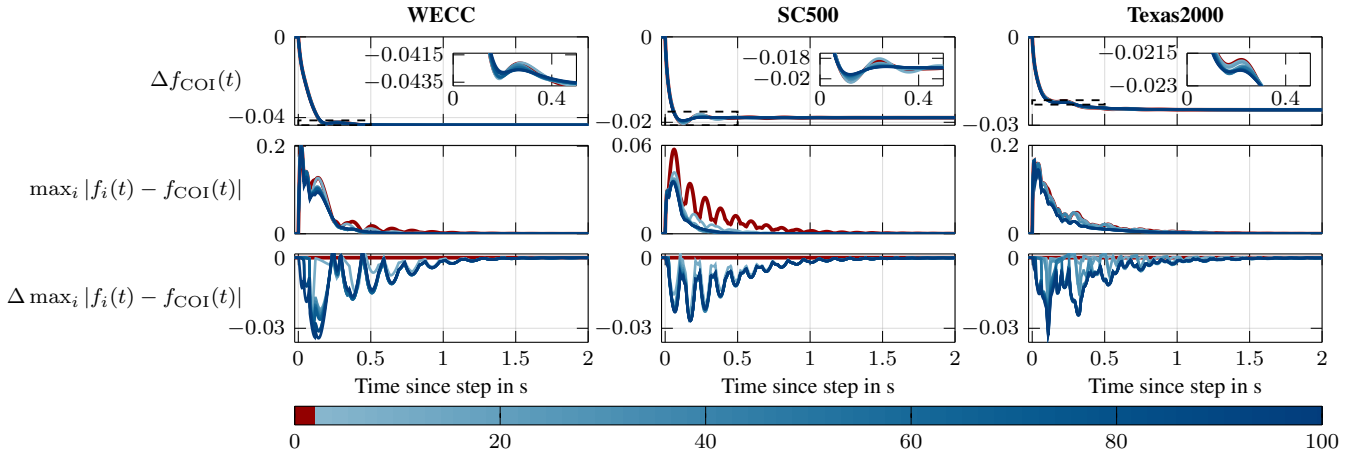


Fig. 7 COI frequency envelope across the activation sweep on WECC, SC500 and Texas2000, shaded by the share of units damped in % (colour bar, the 0% baseline in red). The rows give (in Hz from top downwards) the COI deviation $\Delta f_{\text{COI}}(t)$, the peak relative-COI envelope $\max_i |f_i(t) - f_{\text{COI}}(t)|$ and that envelope minus its 0% baseline.

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